

# BlinkIT Grocery Sales Analysis

## Business Case Study

### Tools Used:

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL
- Power BI
- Excel

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## Business Scenario

### Background

BlinkIT is one of India's leading quick-commerce grocery delivery platforms, serving customers through multiple outlets across different city tiers.

The management wants to understand:

- Which products drive maximum revenue
  - Which outlets perform best
  - How customer ratings impact sales
  - Which areas need operational improvements
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## Business Objectives

### Objectives

#### Increase Revenue

- Improve overall sales performance

#### Optimize Inventory

- Focus on high-demand products

#### Improve Customer Satisfaction

- Analyze customer ratings

## Support Expansion Decisions

- Identify successful outlet formats and locations
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# Dataset Overview

## Dataset Information

**Records:** 8,523

### Key Fields

- Item Type
  - Sales
  - Rating
  - Item Fat Content
  - Outlet Type
  - Outlet Size
  - Outlet Location Type
  - Item Visibility
  - Outlet Establishment Year
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# Key Performance Indicators (KPIs)

## Business KPIs

### Sales KPIs

- Total Sales
- Average Sales

### Customer KPIs

- Average Rating

### Outlet KPIs

- Sales by Outlet Type
- Sales by Outlet Size
- Sales by Location Tier

## Product KPIs

- Sales by Item Category
  - Sales by Fat Content
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# Data Cleaning Process

## Data Preparation

### Performed

- ✓ Checked Missing Values
- ✓ Standardized Item Fat Content
- ✓ Validated Sales Data
- ✓ Removed Duplicates
- ✓ Verified Data Types

### Result

Improved data quality for accurate analysis and reporting.

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# Product Performance Analysis

## Key Findings

### Top Revenue Categories

- Fruits & Vegetables
- Snack Foods
- Household Items
- Frozen Foods
- Dairy Products

### Business Insight

A small number of product categories contribute a significant portion of total revenue.

## **Recommendation**

Increase inventory availability for top-performing categories.

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# **Sales by Fat Content**

## **Analysis**

### **Categories**

- Low Fat
- Regular

### **Findings**

- Regular products generate a larger share of sales.
- Low-fat products continue to show strong demand.

## **Recommendation**

Maintain balanced inventory based on customer preferences.

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# **Outlet Performance Analysis**

## **Outlet Type Comparison**

### **High Performing**

- Supermarket Type 1
- Supermarket Type 2

### **Lower Performing**

- Small Grocery Stores

### **Insight**

Larger retail formats generate significantly higher revenue.

## **Recommendation**

Focus expansion on successful outlet models.

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# Location Analysis

## Sales by Location Tier

### Tier 1

Urban Markets

### Tier 2

Growing Markets

### Tier 3

Emerging Markets

### Key Insight

Tier 3 locations contribute strong sales performance despite smaller populations.

### Recommendation

Explore expansion opportunities in high-performing Tier 3 regions.

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# Customer Rating Analysis

## Customer Satisfaction

### Average Rating Analysis

Higher-rated products generally generate stronger sales performance.

### Benefits

- Increased customer loyalty
- Better repeat purchases
- Higher product demand

### Recommendation

Improve product quality and customer experience.

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# Correlation Analysis

## Relationship Between Variables

### Variables Studied

- Sales
- Rating
- Item Visibility
- Item Weight

### Key Findings

- Product visibility influences sales.
  - Ratings have a positive impact on revenue.
  - Product weight has limited influence on purchasing behavior.
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## Key Insights

### Major Findings

1. Top product categories drive most revenue.
  2. Supermarkets outperform grocery stores.
  3. Tier 3 locations show strong growth potential.
  4. Customer ratings positively impact sales.
  5. Product visibility affects performance.
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## Recommendations

### Quick Wins

- Increase stock of best-selling products.
- Promote highly-rated products.
- Improve product placement and visibility.

## Long-Term Strategies

- Expand successful outlet formats.
  - Use predictive inventory planning.
  - Enhance customer experience initiatives.
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## Business Impact

### Expected Benefits

#### Revenue Growth

10–15% increase in sales potential.

#### Better Inventory Management

Reduced stock shortages and wastage.

#### Improved Customer Satisfaction

Higher ratings and repeat purchases.

#### Strategic Expansion

Data-driven outlet investment decisions.

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## Thank You

BlinkIT Grocery Sales Analysis